Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The harmonic mean of two positive numbers a and b is 4, their arithmeitc mean is A and the geometric mean is G. If $2A + G^2 = 27$, $a + b = \alpha$ and $a - b = \beta$, then the value of $\frac{\alpha}{\beta}$ is equal to

(A) 1

(B) 3

(C) $\frac{5}{2}$

(D) 5

Q.2

Given that α, γ are root of the equation $\mathrm{Ax}^2 - 4\mathrm{x} + 1 = 0$. and β, δ the roots of the equation $\mathrm{Bx}^2 - 6\mathrm{x} + 1 = 0$, the values of A and B such that
α, β, γ and δ are in $\mathrm{H.P.}$ are

(A) $\mathrm{A} = 3,\ \mathrm{B} = 8$

(B) $A = -3, B = 8$

(C) $A = 3, B = -8$

(D) None of these

Q.3

If a, b and c are in H. P. then the value of
$\left(\frac{1}{b} + \frac{1}{c} - \frac{1}{a}\right)\left(\frac{1}{c} + \frac{1}{a} - \frac{1}{b}\right)$ is

(A) $\frac{2}{bc} + \frac{1}{b^2}$

(B) $\frac{3}{c^2} + \frac{2}{ca}$

(C) $\frac{3}{b^2} - \frac{2}{ab}$

(D) None of these

Q.4

If the coefficients of r th, $(r+1)$ th, and $(r+2)$ th terms in the the binomial expansion of $(1+y)^m$ are in A.P., then m and r satisfy the equation

(A) $m^2 - m(4r-1) + 4r^2 - 2 = 0$

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(B) $m^2 - m(4r+1) + 4r^2 + 2 = 0$

(C) $m^2 - m(4r+1) + 4r^2 - 2 = 0$

(D) $m^2 - m(4r-1) + 4r^2 + 2 = 0$

Q.5

If a, b, and c are in A.P., and p and p' are, respectively, A.M. and G.M. between a and b while q, q' are, respectively, the A,M. and G. M. between b and c, then

(A) $p^2 + q^2 = p'^2 + q'^2$

(B) $pq = p'q'$

(C) $p^2 - q^2 = p'^2 - q'^2$

(D) None of these

Q.6

If $\frac{1}{\sqrt{b}+\sqrt{c}}, \frac{1}{\sqrt{c}+\sqrt{a}}, \frac{1}{\sqrt{a}+\sqrt{b}}$ are in A.P. then
$9^{ax+1}, 9^{bx+1}, 9^{cx+1}, x \neq 0$ are in:

(A) GP.

(B) G.P. only if $x < 0$

(C) G.P. only if $x > 0$

(D) none of these

Q.7

The value of $x + y + z$ is $15. If a, x, y, z, b$ are in A.P.
while the value of $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ is $\frac{5}{3}$. If a, x, y, z, b are in H.P. Then the value of a and b are

(A) 2 and 8

(B) 1 and 9

(C) 3 and 7

(D) None

Q.8

If the positive integers a, b, c, d are in AP, then the numbers abc, abd, acd, bcd are in

(A) HP

(B) AP

(C) GP

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(D) None of the above

Q.9

a, b, c are three distinct real numbers and they are in a G.P. if $a+b+c=xb$, then

(A) $x \leq -1$ or $x \geq 3$

(B) $x < -1$ or $x > 3$

(C) $x \leq -1$ or $x > 3$

(D) $x < -3$ or $x > 2$

Q.10

The expression $\frac{a^{n+1}+b^{n+1}}{a^n+b^n}$ is $[a \neq b \neq 0]$ is
(where a and b are unequal non-zero numbers)

(A) A.M. between a and b if $n=-1$

(B) G.M. between a and b if $n=-\frac{1}{2}$

(C) H.M. between a and b if $n=0$

(D) all are correct

Q.11

The equation $\left(a^2+b^2\right)x^2-2b(a+c)x+\left(b^2+c^2\right)=0$ has equal roots. Which one of the following is correct about a, b, and c ?

(A) They are in AP

(B) They are in GP

(C) They are in HP

(D) They are neither in AP, nor in GP, nor in HP

Q.12

Suppose a, b, c are in A.P. and a^2, b^2, c^2 are in G.P. If $a<b<c$ and $a+b+c=\frac{3}{2}$, then the value of a is?

(A) $\frac{1}{2\sqrt{2}}$

(B) $\frac{1}{2\sqrt{3}}$

(C) $\frac{1}{2}+\frac{1}{\sqrt{2}}$

(D) $\frac{1}{2}-\frac{1}{\sqrt{2}}$

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Q.13

If one A.M. A and two G.M.s p and q are inserted between two given numbers, then $\frac{p^2}{q} + \frac{q^2}{p}$ is equal to

(A) A

(B) $2A$

(C) $3A$

(D) $4A$

Q.14

A person purchased one $\mathbf{kg}$ of potatoes from each of 4 places at the rate of $1\ \text{kg}, 2\ \text{kg}, 3\ \text{kg}$ and $4\ \text{kg}$ per rupee respectively. If he has purchased $x\ \text{kg}$ of potatoes per rupee, then x is

(A) 1.92

(B) 2.08

(C) 2.10

(D) none of these

Q.15

If $4a^2 + 9b^2 + 16c^2 = 2(3ab + 6bc + 4ca)$, where a, b, c are nonzero numbers, then a, b and c are in

(A) A.P.

(B) G.P.

(C) H.P.

(D) none of these

Answers & Solutions

Q.1 Answer:

3

Solution:

Given, harmonic mean $H = 4$

We know that, $G^2 = AH$

Since, $2A + G^2 = 27$

$\Rightarrow 2A + AH = 27$

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$\Rightarrow 2A + 4A = 27$

$\Rightarrow A = \frac{27}{6} = \frac{9}{2} = \frac{a+b}{2} \Rightarrow a + b = 9$

$G^2 = AH = \frac{9}{2} \times 4 = 18$

$\Rightarrow ab = 18$

$a - b = \sqrt{a + b^2 - 4ab} = \sqrt{81 - 4 \times 18} = 3$

$\Rightarrow \alpha = 9, \quad \beta = 3$

$\Rightarrow \frac{\alpha}{\beta} = 3$

Q.2 Answer:

$\mathrm{A} = 3, \ \mathrm{B} = 8$

Solution:

α, β, γ **and** δ **are in H.P.** $\Rightarrow \frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}, \frac{1}{\delta}$
are in A.P. Let d **be the common difference of this A.P. Now,** α, γ **are roots of** $Ax^2 - 4x + 1 = 0$
- $\frac{\alpha+\gamma}{a\gamma} = \frac{4/\mathrm{A}}{1/\mathrm{A}} = 4$ **or** $\frac{1}{\alpha} + \frac{1}{\gamma} = 4$ **i.e**
$\frac{1}{\alpha} + \frac{1}{\alpha} + 2\,\mathrm{d} = 4$
or $\frac{1}{\alpha} + \mathrm{d} = 2$ **...(i)**
β, δ **are roots of** $\mathrm{Bx}^2 - 6\mathrm{x} + 1 = 0$
$\therefore \frac{\beta+\delta}{\beta\delta} = \frac{1}{\beta} + \frac{1}{\delta} = \frac{6/\mathrm{B}}{1/\mathrm{B}} = 6$ **or**
$\frac{1}{\alpha} + d + \frac{1}{\alpha} + 3d = 6$
$\frac{1}{\alpha} + 2\,\mathrm{d} = 3$ **...(ii)**
From (i) and (ii), on solving, we get $\frac{1}{\alpha} = 1, \ \mathrm{d} = 1. \therefore \frac{1}{\alpha} = 1, \frac{1}{\beta} = 2, \frac{1}{\gamma} = 3, \frac{1}{\delta} = 4$
Since, $\frac{1}{\alpha\gamma} = \mathrm{A} \therefore \mathrm{A} = 3.$ **Also,** $\frac{1}{\beta\delta} = \mathrm{B}, \therefore \mathrm{B} = 8$
Hence $\mathrm{A} = 3$ **and** $\mathrm{B} = 8$

Q.3 Answer:

$\frac{3}{b^2} - \frac{2}{ab}$

Solution:

Let a, b **and** c **are in H.P.** $\therefore \frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ **arein A. P.**
$\frac{1}{b} - \frac{1}{a} = \frac{1}{c} - \frac{1}{b}$
$\Rightarrow \quad \frac{1}{c} = \frac{2}{b} - \frac{1}{a}$
Consider $\left(\frac{1}{b} + \frac{1}{c} - \frac{1}{a}\right)\left(\frac{1}{c} + \frac{1}{a} - \frac{1}{b}\right)$

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$= \left(\frac{1}{b}+\frac{2}{b}-\frac{1}{a}-\frac{1}{a}\right)\left(\frac{2}{b}-\frac{1}{b}\right)$

Using $\frac{1}{a}+\frac{1}{c}=\frac{2}{b}=\left(\frac{3}{b}-\frac{2}{a}\right)\left(\frac{1}{b}\right)=\frac{3}{b^2}-\frac{2}{ab}$

Q.4 Answer:

$m^2-m(4r+1)+4r^2-2=0$

Solution:

Given ${}^mC_{r-1}, {}^mC_r, {}^mC_{r+1}$ **are in A.P.**

$2{}^mC_r = {}^mC_{r-1}+{}^mC_{r+1}$

$\Rightarrow 2=\frac{{}^mC_{r-1}}{{}^mC_r}+\frac{{}^mC_{r+1}}{{}^mC_r}=\frac{r}{m-r+1}+\frac{m-r}{r+1}$

$\Rightarrow \quad m^2-m(4r+1)+4r^2-2=0.$

Q.5 Answer:

$p^2-q^2=p'^2-q'^2$

Solution:

$2\,\mathrm{b}=\mathrm{a}+\mathrm{c}$; a, p, b, q, c, **are in A.P. Hence,** $p=\frac{a+b}{2}$ **and** $q=\frac{b+c}{2}$

Again , a, p', b, q' and c = are in G.P. Hence, $\mathrm{p}'=\sqrt{ab}$ **and** $\mathrm{q}'=\sqrt{bc}$

$\mathrm{p}^2-q^2=\frac{(a-c)(a+c+2b)}{4}$

$=\frac{(\mathrm{a}-\mathrm{c})(2\,\mathrm{b}+2\,\mathrm{b})}{4} \quad [\because \mathrm{a}+\mathrm{c}=2\,\mathrm{b}]$

$=(a-c)b=ab-bc=p'^2-q'^2$

Q.6 Answer:

GP.

Solution:

$\frac{2}{\sqrt{\mathrm{c}}+\sqrt{\mathrm{a}}}=\frac{1}{\sqrt{\mathrm{b}}+\sqrt{\mathrm{c}}}+\frac{1}{\sqrt{\mathrm{a}}+\sqrt{\mathrm{b}}}$

$=\frac{2\sqrt{b}+\sqrt{a}+\sqrt{c}}{(\sqrt{b}+\sqrt{c})(\sqrt{a}+\sqrt{b})}$

$\Rightarrow 2\sqrt{ab}+2b+2\sqrt{ac}+2\sqrt{bc}$

$=2\sqrt{bc}+2\sqrt{ac}+c+2\sqrt{ab}+a \Rightarrow 2b=a+c$

$\therefore$ a, b, c, **arein A.P.**

$\Rightarrow$ ax, bx, cx, **are in A.P.**

$\Rightarrow$ ax + 1, bx + 1, cx + 1, **are in A.P.**

$\Rightarrow 9^{\mathrm{ax}+1}, 9^{\mathrm{bx}+1}, 9^{\mathrm{cx}+1}$ **are in G.P.**

[See the properites of A.P \& G.P.]

Q.7 Answer:

1 and 9

Solution:

As x, y, z, **are A.** M. **of** a **and** b $\therefore x+y+z=3\left(\frac{a+b}{2}\right)$

$\therefore 15=\frac{3}{2}(\mathrm{a}+\mathrm{b}) \Rightarrow \mathrm{a}+\mathrm{b}=10$ **...(1)**

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Again $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ are A.M. of $\frac{1}{a}$ and $\frac{1}{b}$

$\therefore \frac{1}{x}+\frac{1}{y}+\frac{1}{z}=\frac{3}{2}\left(\frac{1}{a}+\frac{1}{b}\right)$

$\therefore \frac{5}{3}=\frac{3}{2} \cdot \frac{a+b}{ab}$

$\Rightarrow \frac{10}{9}=\frac{10}{ab} \Rightarrow ab=9$ **...(2)**

Solving (1) **and** (2)**, we get** $\mathrm{a}=9,1,\ \mathrm{b}=1,9$

Q.8 Answer:

HP

Solution:

Given, $\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{d}$ **are in A.P.**

$\Rightarrow \frac{1}{a}, \frac{1}{b}, \frac{1}{c}, \frac{1}{d}$ **are in H.P.**

$\Rightarrow \frac{1}{\mathrm{d}}, \frac{1}{\mathrm{c}}, \frac{1}{\mathrm{b}}, \frac{1}{\mathrm{a}}$ **are also in H.P.**

Now, multiply each term by abcd. $\frac{abcd}{d}, \frac{abcd}{c}, \frac{abcd}{b}, \frac{abcd}{a}$ **abc, abd, acd, bcd, are in H.P.**

Q.9 Answer:

$x<-1$ or $x>3$

Solution:

Let $b=ar$ **and** $c=ar^2$**, so that** a, b, c **are in G.P.** $\therefore \mathrm{a}+\mathrm{b}+\mathrm{c}=\mathrm{xb}$

$\Rightarrow \mathrm{a}+\mathrm{ar}+\mathrm{ar}^2=\mathrm{x.ar} \Rightarrow \mathrm{r}^2+(1-\mathrm{x})\mathrm{r}+1=0$

If ris real, then discriminant of $(1) \geq 0$

$\Rightarrow \quad (1-x)^2-4.1.1 \geq 0 \Rightarrow x^2-2x-3 \geq 0$

$\Rightarrow \quad (x+1)(x-3) \geq 0 \Rightarrow x \leq -1$ **or** $x \geq 3$

Now for $x=3$ **we get** $r=1$**, which will make** $a=b=c$ **Also for** $x=-1$**, we get** $r=-1$**, for which** $\mathrm{a}=\mathrm{c}$**, thus** $\mathrm{x}<-1$ **or** $\mathrm{x}>3$

Q.10 Answer:

G.M. between a and b if $n=-\frac{1}{2}$

Solution:

Let $\frac{a^{n+1}+b^{n+1}}{a^n+b^n}=\frac{a+b}{2}$

$\Rightarrow 2a^{n+1}+2b^{n+1}=a^{n+1}+b^{n+1}+ab^n+ba^n$

$\Rightarrow a^{n+1}-a^nb+b^{n+1}-ab^n=0$

$\Rightarrow (a-b)\left(a^n-b^n\right)=0$

$\Rightarrow a^n=b^n$ **it is possible for unequal numbers a and** b **if** $\mathrm{n}=0$

Let $\frac{a^{n+1}+b^{n+1}}{a^n+b^n}=\sqrt{ab}$

$\Rightarrow \mathrm{a}^{\mathrm{n}+1}+\mathrm{b}^{\mathrm{n}+1}=\mathrm{a}^{\mathrm{n}+\frac{1}{2}}\ \mathrm{b}^{\frac{1}{2}}+\mathrm{a}^{\frac{1}{2}}\ \mathrm{b}^{\mathrm{n}+\frac{1}{2}}$

$\Rightarrow \left(a^{n+\frac{1}{2}}-b^{n+\frac{1}{2}}\right)(\sqrt{a}-\sqrt{b})=0$

$\Rightarrow a^{n+\frac{1}{2}}-b^{n+\frac{1}{2}}=0$

which holds true if $n+\frac{1}{2}=0 \Rightarrow n=-\frac{1}{2}$

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Let $\frac{a^{n+1}+b^{n+1}}{a^n+b^n} = \frac{2ab}{a+b}$

$\Rightarrow a^{n+2} + a^{n+1}b + ab^{n+1} + b = 2a^{n+2}a^{n+1}b + 2ab^{n+1}$

$\Rightarrow (a-b)\left(a^{n+1} - b^{n+1}\right) = 0$

$\Rightarrow \mathrm{a}^{\mathrm{n}+1} - \mathrm{b}^{\mathrm{n}+1} = 0 \Rightarrow \mathrm{n} = -1$

Q.11 Answer:

They are in GP

Solution:

The given equation $\left(a^2+b^2\right)x^2 - 2b(a+c)x + \left(b^2+c^2\right) = 0$

has equal roots, so, discriminant $= 0$ **Hence,** $\left\{2b(a+c)^2 - 4\left(a^2+b^2\right)\left(b^2+c^2\right) = 0\right.$

$\Rightarrow 4b^2\left(a^2+c^2+2ca\right) - 4\left(a^2b^2+a^2c^2+b^4+b^2c^2\right) = 0$

$\Rightarrow b^2a^2 + b^2c^2 + 2b^2ca - a^2b^2 - a^2c^2$

$-b^4 - b^2c^2 = 0$

$\Rightarrow \quad 2b^2ca = b^4 + a^2c^2$

$\Rightarrow \quad b4 - 2b^2ca + a^2c^2 = 0$

$\Rightarrow \quad \left(b^2 - ac\right)^2 = 0$

$\Rightarrow \quad b^2 = ac$

$\Rightarrow \quad \mathrm{a, b, c}$ **are in GP.**

Q.12 Answer:

$\frac{1}{2} - \frac{1}{\sqrt{2}}$

Solution:

$a = m - d \quad b = m \quad c = m + d$

d **is the common difference**

$$a + b + c = \frac{3}{2}$$

$$m - d + m + m + d = \frac{3}{2}$$

$$3m = \frac{3}{2}$$

$$m = \frac{1}{2}$$

$a^2, b^2, c^2 \rightarrow G \cdot P$

$b^4 = a^2c^2$

$(m)^4 = (m-d)^2(m+d)^2$

$m^4 = \left(m^2 - d^2\right)^2$

$m^4 = m^4 + d^4 - 2m^2d^2$

$j^4 - 2m^2j^2 = 0$

$d^2\left(d^2 - 2m^2\right) = 0$

$d^2 - 2m^2 = 0$

$d^2 - 2m^2$

$j^2 = 2 \times \frac{1}{4}$

$d^2 = \frac{1}{2}v^2 =$

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$d = 1/\sqrt{2}$

$a = m - d$

$a = \frac{1}{2} - \frac{1}{\sqrt{2}}$

Q.13 Answer:

$2A$

Solution:

Let the two numbers be a and b. Then, $A = \frac{a+b}{2}$

Also, a, p, q, b are in G.P.

$\therefore \quad \frac{p}{a} = \frac{q}{p} = \frac{b}{q}; \therefore \frac{p^2}{q} = a$ **and** $\frac{q^2}{p} = b$

$\therefore \quad \frac{p^2}{q} + \frac{q^2}{p} = a + b = 2A$

Q.14 Answer:

1.92

Solution:

Since we are given the rate per rupee, we use H.M.

$\text{H.M} = \frac{4}{\frac{1}{1}+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}} = 1.92$ **kg per rupee.**

Q.15 Answer:

H.P.

Solution:

$(2a)^2 + (3b)^2 + (4c)^2 - (2a)(3b) - (3b)(4c) - (4c)(2a) = 0$

$\Rightarrow \quad (2a - 3b)^2 + (3b - 4c)^2 + (4c - 2a)^2 = 0$

$\Rightarrow \quad 2a = 3b = 4c \Rightarrow \frac{a}{1/2} = \frac{b}{1/3} = \frac{c}{1/4}$

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